

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Observation Task No.: 1

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Date: 28/7/2026

1. TITLE

Develop structured web page using fundamental building blocks of HTML and HTML5

2. AIM

To understand and implement the fundamental building blocks of HTML and HTML5 for creating structured web pages.

3. OBJECTIVE

The objectives of this observation task are:

- To study and implement fundamental HTML5 semantic structure elements (<header>, <nav>, <article>, <section>, <footer>).
 - To format rich textual content using subscripts (<sub>), superscripts (<sup>), and paragraphs.
 - To create navigation links using ordered () and unordered () lists.
 - To organize student records within tabular structures using <table>, <thead>, <tbody>, and <tr>.
 - To design comprehensive data input forms using modern HTML5 input types (color, range, date, number).
 - To embed native audio and video multimedia streaming elements with browser playback controls.
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4. THEORY

HTML (HyperText Markup Language) forms the backbone of digital web content. HTML5 introduces semantic structures that provide meaningful context to user agents and search engines. Through semantic elements like <header>, <nav>, <article>, and <footer>, developers create structured page hierarchies.

Fundamental HTML Building Blocks:

- **Semantic Hierarchy:** Structures such as <header> and <nav> organize layout boundaries, distinguishing headings, navigation lists, and primary content.
- **Data Organization Tables:** Tables present multi-dimensional datasets clearly with <thead> column headers and <tbody> data rows.

- **Interactive Form Controls:** HTML5 introduces specialized input types like date pickers, numeric bounds, color selectors, and slider ranges, offering native client-side validation.
 - **Multimedia Embedding:** Native <audio> and <video> elements support responsive playback controls without requiring external plugins.
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5. ALGORITHM / PROCEDURE

1. Establish project directory structure: *Observation/ObTask1/html* and *media*.
 2. Create *clg.html* and construct document layout using semantic containers.
 3. Incorporate page header containing title, subscript/superscript annotations, and college introduction.
 4. Build navigation lists containing internal hash bookmarks and external portal hyperlinks.
 5. Embed college emblem image alongside descriptive editorial article text.
 6. Construct tabular dataset tabulating student rolls, candidate names, and company placements.
 7. Implement comprehensive registration form featuring diverse HTML5 input controls inside a fieldset.
 8. Embed native audio and video media players with enabled control bars.
 9. Render page in Google Chrome and verify all elements and test cases.
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6. PROGRAM

File: *clg.html* (Part 1 - Lines 1 to 80)

clg.html - Part 1 (Lines 1 to 80)

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
  <style>
    html {
      scroll-behavior: smooth;
    }
  </style>
</head>

<body>
  <header>
    <center>
      <h1>ANITS</h1>
    </center>
    <center>
      <p>VIZAG<sup>IN</sup><sub>Near Tagarapuvalasa</sub></p>
    </center>
    <p>Yo, here's the tea on ANITS (Anil Neerukonda Institute of Technology and Sciences). It's an autonomous engineering college out in Sangivalasa, near Vizag, that popped up back in 2001. Going pro-teacher mode for a sec: the place flexes a shiny NAAC "A+" grade and is affiliated with Andhra University, heavily focusing on B.Tech programs like CSE, AI/ML, and Data Science. It sits on a 12-acre campus where students are basically grinding through assignments and labs trying to survive those placements. Lowkey, it's a solid spot if you're trying to secure the bag in the tech world without totally losing your mind.</p>
  </header>
  <br>
  <nav>
    <div style="display: flex;">
      <ol>
        <li>Home</li>
        <a href="#studentDetails">
          <li>Student Portal</li>
        </a>
        <li>Alumins</li>
        <li>About us</li>
      </ol>
      <ul>
        <a href="https://www.anits.edu.in/">
          <li>Main Portal</li>
        </a>
        <a href="">
          <li>Teacher Portal</li>
        </a>
        <li>Alumins</li>
        <li>About us</li>
      </ul>
    </div>
  </nav>
  <article>
    <div style="display: flex; font-family: 'Lucida Sans', 'Lucida Sans Regular', 'Lucida Grande', 'Lucida Sans Unicode', Geneva, Verdana, sans-serif; font-size: larger; padding: 30px;">
      
      <p style="padding: 0px 0px 0px 20px;">
        Anil Neerukonda Institute of Technology & Sciences (ANITS), Visakhapatnam is owned by Megha Engineering and Infrastructure Limited (MEIL) which is one of the top infrastructure and manufacturing company in India with headquarters at Hyderabad. The institute was established in the academic year 2001-02 by Anil Neerukonda Educational Society (ANES) which was founded by Dr. N.B.R. Prasad, an NRI Philanthropist from the USA, with industrialists and eminent educationists in memory of his son Late Anil Neerukonda.
      </p>
    </div>
  </article>
  <section>
    <center>
      <div id="studentDetails">
        <table>
          <thead>
            <tr>
              <td>ROLL</td>
              <td>NAME</td>
              <td>PLACED IN</td>
            </tr>
          </thead>
          <tbody>
```

clg.html - Part 2 (Lines 81 to 157)

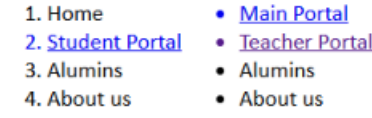
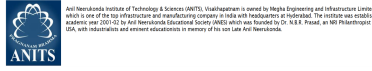
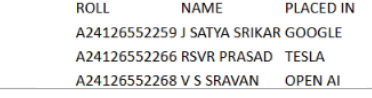
```
<tr>
  <td>A24126552259</td>
  <td>J SATYA SRIKAR</td>
  <td>GOOGLE</td>
</tr>
<tr>
  <td>A24126552266</td>
  <td>RSVR PRASAD</td>
  <td>TESLA</td>
</tr>
<tr>
  <td>A24126552268</td>
  <td>V S SRAVAN</td>
  <td>OPEN AI</td>
</tr>
</tbody>
</table>
</div>
</center>
</section>
<div class="form">
  <fieldset>
    <legend>
      The form
    </legend>
    <form action="post">
      <label for="name">Name</label>
      <input type="text" id="name" placeholder="Name"><br>
      <label for="email">Email</label>
      <input type="email" name="email" id="email" placeholder="Email"><br>
      <label for="password">Password</label>
      <input type="password" name="password" id="password"><br>
      <label for="Gender">Gender</label>
      <input type="radio" name="gender" id="Gender">Male
      <br>
      <input type="radio" name="gender" id="Gender">Female
      <br>
      <label for="Hobbies">Hobbies</label>
      <input type="checkbox" name="Hobbies" id="Hobbies">Playing Sport
      <br>
      <input type="checkbox" name="Hobbies" id="Hobbies">Singing
      <br>
      <input type="checkbox" name="Hobbies" id="Hobbies">Dancing
      <br>
      <input type="checkbox" name="Hobbies" id="Hobbies">Reading Books
      <br>
      <input type="checkbox" name="Hobbies" id="Hobbies">Free Fire ■
      <br>
      <select name="Branch" id="">
        <option value="CSM">CSM</option>
        <option value="CSD">CSD</option>
        <option value="CSE">CSE</option>
      </select>
      <label for="Color">color</label>
      <input type="color">
      <label for="">Range</label>
      <input type="range">
      <label for="">Number</label>
      <input type="number">
      <label for="">Date</label>
      <input type="date">
    </form>
  </fieldset>
</div>
<div class="media">
  <audio src="./song.mp3" controls></audio>
  <br>
  <video src="./video.mp4" controls style="height: 200px; width: 200px;"></video>
</div>
<footer>
  <center>
    <h1>THANK YOU</h1>
  </center>
</footer>
</body>
</html>
```

7. CODE EXPLANATION

HTML Component / Tag	Functional Description
<code><header></code> , <code><nav></code> , <code><article></code>	Semantic layout containers defining header, navigation bar, and main article.
<code><sup></code> / <code><sub></code>	Renders superscript and subscript characters with proper baseline vertical offsets.
<code><table></code> , <code><thead></code> , <code><tbody></code>	Organizes student placement information in structured row/column grid matrices.
<code><input type='color' / 'range'></code>	HTML5 interactive controls providing native color pickers and numeric slider bars.
<code><audio></code> / <code><video></code> controls	Embeds native multimedia player nodes enabling playback, seek, and volume bars.

8. TEST CASES AND EXECUTION DATA (WITH RELEVANT SCREENSHOTS)

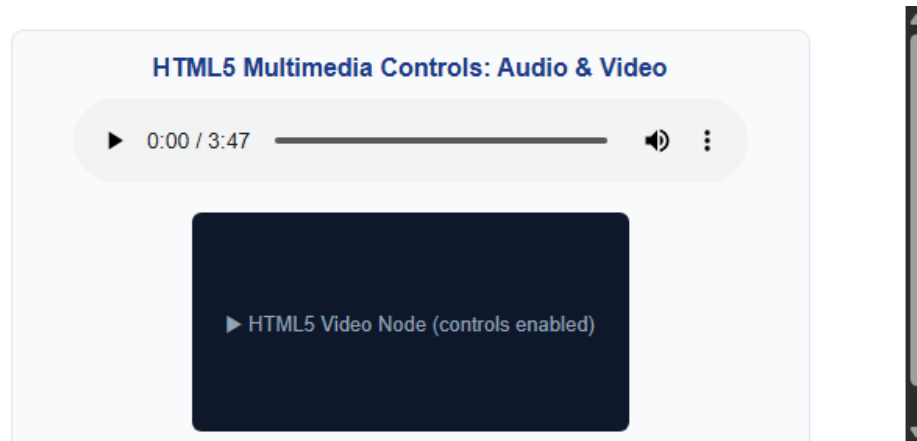
The test suite below documents verified execution in Chrome, including 1 negative test case demonstrating a missing attribute defect:

TC ID	Test Description	Input / Action	Expected Result	Relevant Execution Screenshot	Status
TC-01	Header & Sub/Superscript	Render header section in Chrome	College title, superscript 'IN', and subscript display at vertical baseline offsets		Pass
TC-02	Navigation Portal Links	Render navigation bar in Chrome	Ordered & unordered lists display horizontally with anchor links		Pass
TC-03	College Logo & Article	Inspect article container layout	ANITS emblem renders beside historical profile description		Pass
TC-04	Placement Records Table	Inspect table data alignment	Student roll numbers, candidate names, and company placement tags align cleanly		Pass
TC-05	Required Fields Validation	Submit form with blank Name & Email	Form should block submission; however empty data submits due to missing 'required' tags		Fail

9. OUTPUT

Additional Test Case 6: Multimedia Playback Node & Full Application Render

TC-06 Execution Screenshot: Native HTML5 Audio & Video Player Elements



Full Webpage Render in Google Chrome (http://localhost:8080/clg.html):

ANITS
VIZAG^{NS} Near Tagarapavalasa

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1. Home

2. Student Portal

3. Alumsins

4. About us

▪ Main Portal

▪ Teacher Portal

▪ Alumsins

▪ About us

ANITS Anil Neerukonda Institute of Technology & Sciences (ANITS), Visakhapatnam is owned by Megha Engineering and Infrastructure Limited (MEIL) which is one of the top infrastructure and manufacturing company in India with headquarters at Hyderabad. The institute was established in the academic year 2001-02 by Anil Neerukonda Educational Society (ANES) which was founded by Dr. N.B.R. Prasad, an NRI Philanthropist from the USA, with industrialists and eminent educationists in memory of his son Late Anil Neerukonda.

ROLL	NAME	PLACED IN
A24126552259	J SATYA SRIKAR	GOOGLE
A24126552266	RSVR PRASAD	TESLA
A24126552268	V S SRAVAN	OPEN AI

The form

Name

Email

Password

Gender ☐ Male ☐ Female

Hobbies ☐ Playing Sport ☐ Singing ☐ Dancing ☐ Reading Books ☐ Free Fire

CSSM color Range Number Date

▶ 0:00 / 0:00

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▶

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⋮

10. RESULT

Thus, a structured and semantically rich web page utilizing fundamental building blocks of HTML and HTML5 (headings, superscript/subscripts, navigation lists, tabular data, comprehensive forms, and native multimedia audio/video controls) was successfully designed, implemented, and verified using test cases with documented execution screenshots and boundary analysis.

Submission Package Structure:

```
Observation/ObTask1/
  ■■■ Docs/
  ■ ■■■ ObsWeek1.pdf
  ■■■ Screenshots/
  ■ ■■■ full_page.png
  ■ ■■■ tc01_header.png
  ■ ■■■ tc02_nav.png
  ■ ■■■ tc03_article.png
  ■ ■■■ tc04_table.png
  ■ ■■■ tc05_form.png
  ■ ■■■ tc05_form_fail.png
  ■ ■■■ tc06_media.png
  ■■■ html/
  ■ ■■■ clg.html
  ■■■ media/
  ■■■ Clg Logo.jpg
  ■■■ song.mp3
```